

Multiple Discrete Random Variables

Lecture 5

ตัวแปรสุ่มหลายตัว 1 ตัว แบบไม่ต่อเนื่อง

Topics in this lecture

- Joint PMF
- Conditional PMF
- Conditional expectation
- Independence of random variables

Joint PMF

↳ n. an probability mass function RV ၁၇၇၇၇၇ ၁ မှီ

- Let X and Y be discrete random variables, the joint PMF of X and Y , denoted by $p_{X,Y}(x,y)$, captures the probabilities of the events in the form of $\{X = x, Y = y\}$
- Joint probability mass function

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

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Compute Probability from Joint PMF

- Finding probability of event A

$$P(A) = \sum_{(x,y) \in A} p_{X,Y}(x,y)$$

- Sum over all values of joint pmf

sum over all values of joint pmf

$$\sum_{\forall(x,y)} p_{X,Y}(x,y) = 1$$

Example 5.1

X, Y = ผลลัพธ์ที่ออกมาบนลูกเต๋าสองตัว 4 หน้า (X = ผลลัพธ์ที่ 1 , Y = ผลลัพธ์ที่ 2)

- Let X and Y be the marks that show up after rolling two 4-faced dice. The joint PMF $p_{X,Y}(x,y)$ is given in the following table. Find the probability of an event $\{X > 3, Y < 3\}$

สมมติ joint

		X				
		1	2	3	4	$X > 3$
Y	1	1/24	2/24	2/24	1/24	$Y < 3$
	2	1/24	2/24	2/24	1/24	
	3	1/24	1/24	1/24	1/24	
	4	2/24	2/24	2/24	2/24	

$$P(A) = \sum_{(x,y) \in A} p_{X,Y}(x,y)$$

$$P(\{X > 3, Y < 3\}) = p_{X,Y}(4,1) + p_{X,Y}(4,2)$$

$$P(\{X > 3, Y < 3\}) = \frac{1}{24} + \frac{1}{24} = \frac{2}{24} \neq$$

Example

- A statistic shows that 10% of people entering a store will buy an iPhone, 30% will buy an iPad, and 60% will not buy anything. If 10 people enter the store today, what is the probability that exactly 2 iPhone and 2 iPad will be sold?

Marginal PMF

↳ PMF ของ RV ที่ได้จาก joint PMF ของ 2 RV (joint)

$$\begin{aligned}
 p_X(x) &= P(X = x) \\
 &= P(X = x, Y = \text{anything}) \\
 &= \sum_y p_{X,Y}(x, y)
 \end{aligned}$$

ใน PMF ของ X จาก joint
จะรวม x, y ทั้งหมด

Similarly

$$p_Y(y) = \sum_x p_{X,Y}(x, y)$$

Example 5.2

X, Y = ผลลัพธ์ที่ออกมาบนลูกเต๋าสี่หน้า 2 ลูก (X = ผลลัพธ์ที่ 1 , Y = ผลลัพธ์ที่ 2)

- Let X and Y be the marks that show up after rolling two 4-faced dice. The joint PMF $p_{X,Y}(x,y)$ is given in the following table. Find the marginal PMF of X .

		X			
		1	2	3	4
Y	1	1/24	2/24	2/24	1/24
	2	1/24	2/24	2/24	1/24
	3	1/24	1/24	1/24	1/24
	4	2/24	2/24	2/24	2/24

$$p_X(x) = \sum_y p_{X,Y}(x,y)$$

$$\textcircled{1} \quad p_X(1) = \sum_{y=1}^4 p_{X,Y}(1,y)$$

$$= \frac{1}{24} + \frac{1}{24} + \frac{1}{24} + \frac{2}{24} = \frac{5}{24}$$

$$\textcircled{2} \quad p_X(2) = \sum_{y=1}^4 p_{X,Y}(2,y) = \frac{7}{24}$$

$$\textcircled{3} \quad p_X(3) = \sum_{y=1}^4 p_{X,Y}(3,y) = \frac{7}{24}$$

$$\textcircled{4} \quad p_X(4) = \sum_{y=1}^4 p_{X,Y}(4,y) = \frac{5}{24}$$

$$\therefore p_X(x) = \begin{cases} 5/24, & x = 1, 4 \\ 7/24, & x = 2, 3 \\ 0, & \text{otherwise} \end{cases}$$

Conditional PMF

↳ ปร. หา PMF ของ RV เมื่อรู้เหตุการณ์หนึ่งเกิดขึ้น

- Conditional PMF on Event A

$$p_X(x|A) = P(X = x|A) = \frac{P(\{X = x\} \cap A)}{P(A)}, \quad P(A) > 0$$

- Conditional PMF

$$p_{X|Y}(x|y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

↳ joint pmf
marginal pmf(y)

Example 5.3

X, Y = ผลลัพธ์ที่ออกมาบนลูกเต๋าสี่หน้า 2 ลูก โดยที่ X = ผลลัพธ์ที่ 1, Y = ผลลัพธ์ที่ 2

- Let X and Y be the marks that show up after rolling two 4-faced dice. The joint PMF $p_{X,Y}(x,y)$ is given in the following table. Find the conditional PMF of $X|Y=2$.

$$p_{X|Y=2}(x|y) = \frac{p_{X,Y}(x, 2)}{p_Y(2)}$$

② หา joint

$$\begin{aligned} p_{X|Y=2}(1|y=2) &= \frac{p_{X,Y}(1, 2)}{p_Y(2)} = \frac{1/24}{6/24} = \frac{1}{6} \\ &= \frac{p_{X,Y}(2, 2)}{p_Y(2)} = \frac{2/24}{6/24} = \frac{2}{6} \\ &= \frac{p_{X,Y}(3, 2)}{p_Y(2)} = \frac{2/24}{6/24} = \frac{2}{6} \\ &= \frac{p_{X,Y}(4, 2)}{p_Y(2)} = \frac{1/24}{6/24} = \frac{1}{6} \end{aligned}$$

		X			
		1	2	3	4
Y	1	1/24	2/24	2/24	1/24
	2	1/24	2/24	2/24	1/24
	3	1/24	1/24	1/24	1/24
	4	2/24	2/24	2/24	2/24

① หา $p_Y(y)$
 หาผลรวมของ
 คอลัมน์ = $p_Y(2)$

$$p_{X|Y=2}(x|y) = \begin{cases} 1/6, & x = 1, 4 \\ 2/6, & x = 2, 3 \\ 0, & \text{otherwise} \end{cases}$$

↓
 หา sum ของ joint ที่ $y=2$
 total = 1

Example

- Pick a number X at random from a set $\{1, 2, 3, 4, 5\}$. Then, pick another random number Y , where Y is a positive integer that is less than or equal to X .
 - Find the joint PMF of X, Y
 - Find the conditional PMF of $X \mid Y = 3$

Conditional Expectation

↳ finding $X|Y$

- Conditional expectation of $X|Y = y$

$$E[X|Y = y] = \sum_x x p_{X|Y}(x|y)$$

- Finding $E[X]$ from conditional expectation

$$E[X] = E[E[X|Y]] = \sum_y E[X|Y = y] p_Y(y)$$


Example 5.4

X, Y = ผลลัพธ์ที่ออกมาบนลูกเต๋าสองตัว 4 หน้า (X = ผลลัพธ์ที่ 1, Y = ผลลัพธ์ที่ 2)

- Let X and Y be the marks that show up after rolling two 4-faced dice. The joint PMF $p_{X,Y}(x,y)$ is given in the following table. Find the conditional expectation $E[X|Y=2]$.

Joint PMF

		X			
		1	2	3	4
Y	1	1/24	2/24	2/24	1/24
	2	1/24	2/24	2/24	1/24
	3	1/24	1/24	1/24	1/24
	4	2/24	2/24	2/24	2/24

$$p_{X|Y=2}(x|y) = \begin{cases} 1/6, & x = 1, 4 \\ 2/6, & x = 2, 3 \\ 0, & \text{otherwise} \end{cases}$$

$$E[X|Y=2] = \sum_x x p_{X|Y=2}(x|y) \\ = (1 \times \frac{1}{6}) + (2 \times \frac{2}{6}) + (3 \times \frac{2}{6}) + (4 \times \frac{1}{6})$$

$$E[X|Y=2] = \frac{15}{6} = 2.5$$

Example

- Pick a number X at random from a set $\{1, 2, 3, 4, 5\}$. Then, pick another random number Y , where Y is a positive integer that is less than or equal to X .
- Find the conditional expectation $E[X \mid Y = 3]$

ความน่าจะเป็นอิสระต่อกันของ ตัวแปรสุ่ม 2 ตัว

Independence of 2 RV

- If X and Y are independent discrete random variables, then the joint pmf is equal to the product of the marginal pmf

๗ ถ้าเอ่ทำกัน = เ่้นอิสระต่อกัน

Joint pmf = M pmf_X • M pmf_Y

$$p_{X,Y}(x,y) = p_X(x)p_Y(y) \quad , \quad \forall x,y$$

Example 5.5

X, Y = ผลลัพธ์ที่ออกมาบนลูกเต๋าสี่หน้า 2 ลูก โดยเต๋าสี่หน้า 4 หน้า (X = ผลลัพธ์ที่ 1, Y = ผลลัพธ์ที่ 2)

- Let X and Y be the marks that show up after rolling two 4-faced dice. The joint PMF $p_{X,Y}(x,y)$ is given in the following table. Are X, Y independent?

เงื่อนไข $\rightarrow p_{X,Y}(x,y) = p_X(x) \cdot p_Y(y)$

① $X = 1, Y = 1$

$\rightarrow 1/24 \neq 5/24 \cdot 6/24$

ไม่เป็นอิสระต่อกัน

		X			
		1	2	3	4
Y	1	1/24	2/24	2/24	1/24
	2	1/24	2/24	2/24	1/24
	3	1/24	1/24	1/24	1/24
	4	2/24	2/24	2/24	2/24

$x=1, y=1$
 $p_{X,Y}(1,1) = \frac{1}{24}$
 $p_X(1) = \frac{1}{24} + \frac{1}{24} + \frac{1}{24} + \frac{2}{24} = \frac{5}{24}$
 $p_Y(1) = \frac{1}{24} + \frac{2}{24} + \frac{2}{24} + \frac{1}{24} = \frac{6}{24}$
 $\frac{1}{24} \neq \frac{5}{24} \times \frac{6}{24}$ ✗